



PUBLIC

Optimize the technical runtime of SAP S/4HANA conversion

Using SUMToolBox and TDO app.

Version: 1.0

Date: 2026-09-02

Owner: Ananth Pai

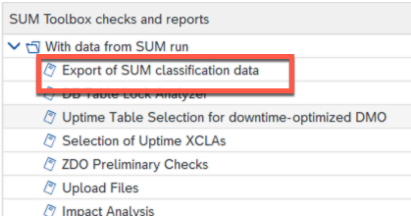
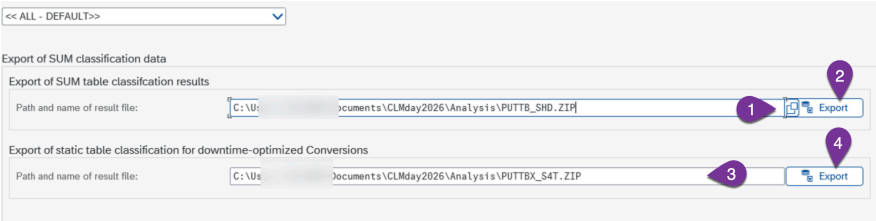
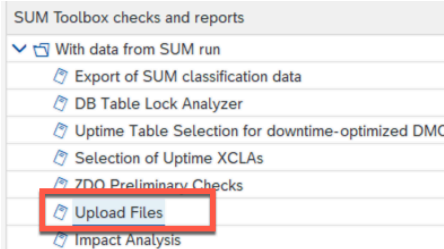
Objective

Once the SAP S/4HANA conversion is completed, there are 2 major things that needs to be identified.

1. Impact of conversion to the Business as usual (BAU)
2. Ways to optimize the overall runtime (Uptime/downtime)

There are 2 tools that will be of use here.

1. SUMToolbox (ABAP based)
2. TDO app (SAP service url – <https://tdo.cloud.sap>)

1	Downloading the table classification data after conversion	Once the SAP S/4HANA conversion is completed, you can download the table classification data. This is needed for you to do further analysis in the SUMtoolbox. This is also needed for the trigger simulation In the system use the TCODE STBX and select the “Export of SUM classification data”  Select the appropriate path and click on the export button 
2	Uploading files for analysis	The PUTTB_SHD.ZIP and ZDIMPANA.ZIP file are needed for analysis of SAP S/4HANA conversion. In STBX tcode select the option upload files 

System Details

System ID: ABA System Type: PRODUCTION

Upload PUTTB SHD Zip File

Table Classification Data: C:\Users\jnts\CLMday2026\Analysis\PUTTB_SHD.ZIP

Remark: PUTTB SHD FILE FROM ABA SYSTEM

Upload IMPANA Zip File

Table Statistics (ZDIMPANA.ZIP): C:\Users\jnts\CLMday2026\Analysis\ZDIMPANA_CLMDemo.ZIP

Remark: ZDIMPANA file from production system

Upload

Upload the PUTTB_SHD.ZIP file that was downloaded from Step 1.

The ZDIMPANA.XML file is provided in the same folder of this exercise.

Provide appropriate understandable description and click upload button. Once the upload has been completed, you should see the entry like below.

Uploaded PUTTB SHD Files

System ID	System Type	Date	Time	Created By	Remark
ABA	PRODUCTION	08/27/2026	11:45:20	10STEPS2S4	CLASISFICATION DATA

Uploaded IMPANA Files

System ID	System Type	Date	Time	Created By	Remark
ABA	PRODUCTION	08/27/2026	11:45:20	10STEPS2S4	IMPACT ANALYSIS UPLOAD

Selection of XCLA's for uptime processing in the subsequent cycles to optimize the downtime

The XCLA's can be moved to uptime, its not done by default. But SUMToolbox gives you an option to analyze the XCLA's and the impacts to the tables its accessing.

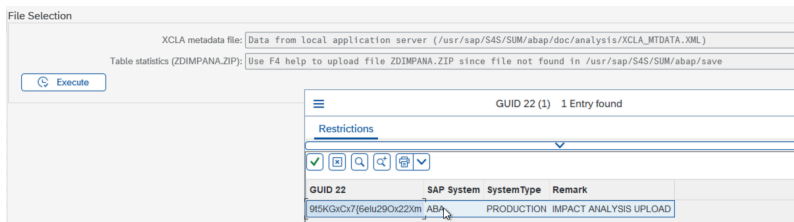
In STBX tcode, select the option "Selection of Uptime XCLA's"

SUM Toolbox checks and reports

- With data from SUM run
 - Export of SUM classification data
 - DB Table Lock Analyzer
 - Uptime Table Selection for downtime-optimized DMO
 - Selection of Uptime XCLAs**
 - ZDO Preliminary Checks

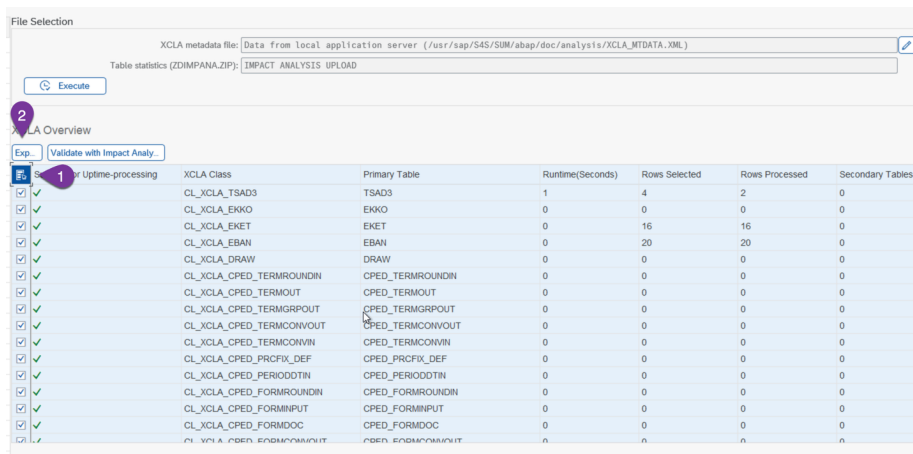
You will need the XCLA_MTDATA.XML file from the complete S/4HANA conversion. This would ideally be in the SUM/abap/doc/analysis/ folder. If you are running the STBX tcode on the application server where the SUM folder access is available to the SAP system, then it would list the path automatically, else you can upload the file.

Select the recently uploaded ZDIMPANA.ZIP file from the list. Ideally this should be from production, mainly from the similar week you plan to run your cutover .



Click Execute button to get the list of allowed XCLA's for uptime processing.

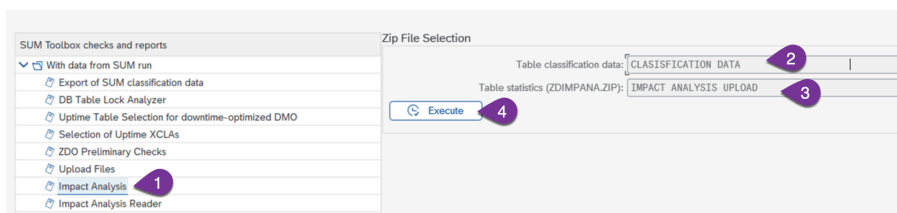
Based on this you can select the relevant XCLA's to uptime



Click the export button, then you will be able to download the XML file which can be provided to SUM for subsequent cycles, which will make the SUM to perform the XCLA's in uptime for the selected XCLA's

Impact analysis

You can generate multiple ZDIMPANA from the production system for different time intervals and check this against the table classification. This will give you a good understanding on which tables would be flagged for read only and based on this you will have to understand what is the **business impact when you are running the SAP S/4HANA conversion** on the production system.



The output looks like below. Please note your production system output would look totally different. The tables that are marked as read only will have impact in production system. Ideally these are customizing tables, and the expectation is that there are no customizing done in the production system while you are running the conversion. Also the tables which get frequent updates will need to be analyzed further.

		Overview Overview SUM scenario: [0000/60C] Source system ID: [ABA] Database platform: [ORACLE] Number of evaluated days: [60] Contains imports: [Yes] Display Periods Overall Summary Estimated DB size for closed tables (GB): [N/A] Change volume for all tables/day (Statements/Rows in mio): [4.3] / [6.8] Total number of closed tables: [0] Estimated growth for logging tables per day (GB): [1] Total number of read-only tables: [27217] Online replication volume for closed tables per day (mio): [3.2] Impact Analysis Findings Read-Only Overview Number of read-only tables found in trace data: [8] Number of large tables: [N/A] with error: [3] with warning: [1] with information: [8] Number of tables with SLT database triggers: [0] Number of tables with Non-SLT database triggers: [0] Table Name Sev. Category Change Statements/Day Row Changes/Day Message Text Package Applic. Component Software Compo ADM_CRT Read-Only 0.1 0.0 TABL ADM_CRT will become read-only, but has 0.1/0.0(statement/row) changes/day BEWU CA-EUR-CHV SAP_ABA TFAV Read-Only 1.5 0.0 TABL TFAV will become read-only, but has 1.5/0.0(statement/row) changes/day FBAS F1 SAP_FH TFAVT Read-Only 1.5 0.0 TABL TFAVT will become read-only, but has 1.5/0.0(statement/row) changes/day FBAS F1 SAP_FH TFBVF Read-Only 81.2 0.0 TABL TFBVF will become read-only, but has 81.2/0.0(statement/row) changes/day FBASCORE F1 SAP_FH
5	Impact analysis reader	This option will allow you to understand the behavior of the tables in production system for the given time interval where the data is recorded. It would give you the details about inserts/ updates/ deletes/ per day. Also gives you information about SLT triggers, which would help in identifying the tables that would need initial loads after the conversion is completed to the side car system. Select the uploaded impact analysis in previous steps and click execute. The below details are about tables, the size, changes etc. File Selection Table statistics (ZDIMPANA.ZIP): [IMPACT ANALYSIS UPLOAD] 1 Execute 2 Overview Overview Source system ID: [ABA] Source system License Number: [0020000070] Database platform: [ORACLE] Export Date: [08/01/2024] Contains imports: [Yes] Number of evaluated days: [61] Display Periods 1 Overall Summary Estimated DB size(GB) of evaluated Tables: [11272.884] Number of Tables with Recorded changes: [54] Number of tables with SLT database triggers: [5] Number of tables with Non-SLT database triggers: [] Insert statements per day (millions): [1] Inserted rows per day (millions): [< 1] Update statements per day (millions): [3] Updated rows per day (millions): [< 1] Delete statements per day (millions): [< 1] Deleted rows per day (millions): [< 1] Tablename Package Software Compone Number of Rows Tabsize in GB Change Statements/Day Changed Rows/Day Inserts/Day Row inserts/Day Updates/Day Row Updates/Day Deletes/Day Row Deletes/Day SLT Triggers DB Triggers ADMJ_ST SARC SAP_BASIS 53,950 0.008 3,177,720.2 0 15.7 0 3,177,704.5 0 0.0 0 0 0 ZD_OUTPL ZPT00 38,796 10,000.012 641,889.3 0 600,000.6 0 641,888.7 0 0.0 0 0 ZD_CTRL ZPT00 21,486,798 2.152 163,105.0 0 162,533.6 0 570.4 0 1.0 0 0 ZD_DELV ZPT00 213,902,704 24.272 148,459.1 0 0.0 0 148,459.1 0 0.0 0 0 TABLE1 ZPT00 1,210,000,257 1,230.012 100,550.0 0 100,000.0 0 500.0 0 50.0 0 3 ZDANALY ZPT00 11,539,016 1.769 24,805.6 0 10,795.2 0 14,010.2 0 0.2 0 0 TRFM VZDC SAP_APPL 1 1.000 20,325.1 0 0.0 0 20,325.1 0 0.0 0 0 ADMJ_RUN SARC SAP_BASIS 54,808 0.005 11,045.0 0 16.9 0 11,028.1 0 0.0 0 0 ZIDOC ZV00 22,468,373 1.671 7,267.0 0 0.0 0 7,267.0 0 0.0 0 0 ZD_OUTMT ZPT00 3,033,771 0.728 4,303.8 0 1,172.6 0 3,131.2 0 0.0 0 0 ADMJ_JOB SARC SAP_BASIS 3,410 0.001 3,601.6 0 3,320.6 0 170.1 0 110.9 0 0 ZD_ABL F ZPT00 1,703,314 0.194 3,317.1 0 0.0 0 3,317.1 0 0.0 0 0
6	TDO app	In this exercise we will see which are the phases/step running longer in uptime and downtime. Open the link https://tdo.cloud.sap Click the button Analyze maintenance events.

Technical Downtime Optimization

Main Menu

Latest Maintenance Events Analyze Maintenance Events

System ID	Host	Scenario	Start Date
Upload UPGANA and APPLANA files			

Upload UPGANA and APPLANA files

Drop files to upload them or use the "Select a file" button.

Select a file

Upload Close

Upload UPGANA and APPLANA files

Please specify the system type and run type for each uploaded file

File Upload: UPGANA.XML System Type: Development Run Type: Other Pending:0%

Select a file

Upload Close

Upload the UPGANA.XML file provided to you in the exercise folder.

Once the upload is completed, you can select the details by searching "all events and filters"

Main Menu

Latest Maintenance Events Analyze Maintenance Events

System ID	Host	Scenario	Start Date
> All events and filters			

Select an Event

System ID:
ABA

Scenario:
1

Installation Number:
4567

Host:

Event Type:

Run Type:

System Type:

Source Release:

Target Release:

System Number:

Start Date:
e.g. 31.12.2026

End Date:
e.g. 31.12.2026

Maintenance Planner ID:

ClearAdapt Filters

Maintenance Events (1)

DownloadUpload

System ID	Host	Scenario	System Type	Start Date	End Date
ABA	vhcalabaci	Downtime-Optimized Conversion to S/4HANA using DMO		8/20/2026, 9:34:29 PM	8/25/2026, 6:38:25 PM

Enter the System ID and search for the right event. Click the system below to get more details.

You can see the details something like below.

Downtime-Optimized Conversion to S/4HANA using DMOSystem Data App

Event	Source	Target
Installation	0123456789 SAP BASIS Release	740 SAP BASIS Release
Type	S/4HANA Conversion	Product Release ECC 6.0 EHP 7
Product	SAP ERP	Start 8/20/2026, 9:34:29 PM
Host	vhcalabaci	Database Sybase
System	00000000850942159	Size 113.47 GiB

Documents (2):
Upgrade analytics documentViewDownloadUpload

DurationsRatingSimulateCutover PlanSystem Info

Major DurationsTimelineTop Contributors

You can click on the blocks to see more details

Overall Runtime

Uptime
[2 days and 12 hours]

Uptime Analysis

Uptime SUM
[2 days and 12 hours]

■ Uptime ■ Technical Downtime ■ Uptime SUM ■ Financial conversion (Uptime)

Total SAP S/4HANA conversionUptimeTechnical Downtime

Explore all the buttons, like Rating, Simulate, Cutover plan, System info.

The TDO app and SUMToolbox gives you a lot of data point which can be used to optimize the overall technical downtime for the SAP S/4HANA conversion.

www.sap.com.